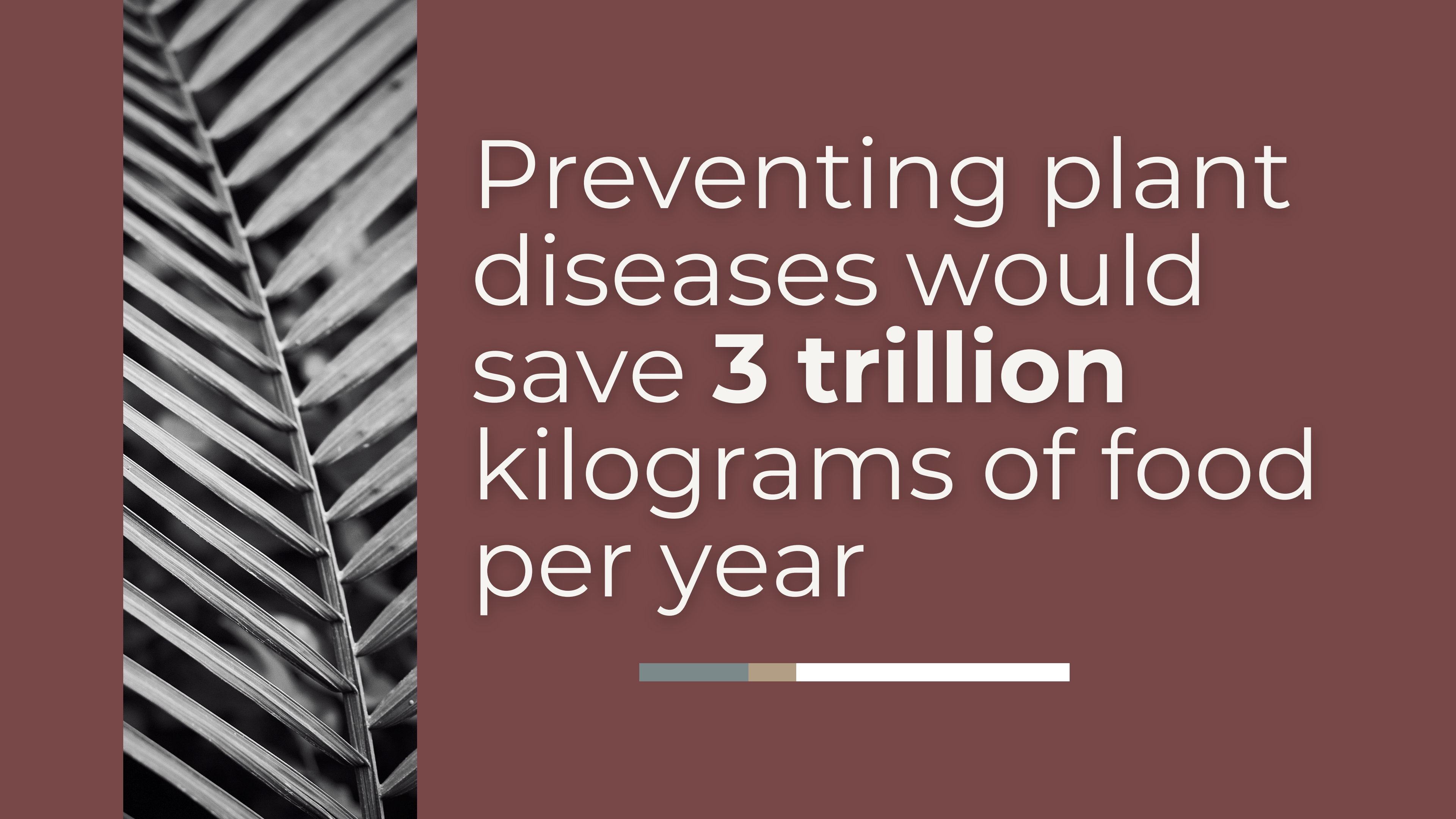


Project Omicron



Kalepu Yashvardhan
Pratham Yadav





Preventing plant
diseases would
save **3 trillion**
kilograms of food
per year

Objective



Our Goal

01 Make a world from Pixels to Plant Health

Be accurate

02 Achieve $\geq 90\%$ validation accuracy with Bayesian confidence

Be for the underprivileged

03 Our Stakeholders are poorest farmers. We give them cutting-edge tech for farming

Help in completing SDG's

04 Reduce 28-30% annual crop losses, contributing to UN SDG Goal 2: Zero Hunger

Our Model

- 01 **Leaf Image Classification Model:** A CNN-based vision model trained to detect three diseases plus healthy leaves with high confidence.
- 02 **Dataset Augmentation Pipeline:** 500+ augmented images per class to boost accuracy, robustness, and real-world generalization, using Orange.
- 03 **Confidence-Based Diagnostics:** Model outputs both disease prediction and confidence score for transparent, reliable decision-making.
- 04 **User-Friendly Assistant Interface:** A simple app/dashboard that lets farmers upload images and receive instant diagnosis and treatment guidance.

Our Solution



But Leaves don't grow on a PPT

Precision Agriculture Assistant

AI-Powered Multi-Crop Disease Diagnosis

📍 Location (Optional)

Delhi



Diagnosis: Cherry - Healthy

Pipeline

Front-End

Input Methods

- Drag and Drop
- File Picker
- Live Camera Capture

Send To Server

- Fetch API
- Async POST request → /predict

Preview and Package

- FileReader → Base64 preview
- FormData → multipart request

Server

Flask Endpoint

- /predict route (POST)
- secure_filename()
- Extension whitelist (.png, .jpg, .jpeg)

File Validation

- PIL img.verify()
- Detects corrupted or invalid images

Image Preprocessing

Conversion of Image

- Resized into size the model can read
- Tensors are converted (uint8 → float32)

Batch Dimension

- Shape: [3,224,224] → [1,3,224,224]
- Each image is tagged and stored in the server

Normalization

- ImageNet mean & standard deviation

AI Inference

Feature Extraction

- Inverted Residual Blocks (17 layers)
- Depth wise Separable Convolutions
- Optimized for low computation

Prediction Logic

- SoftMax → class probabilities
- ArgMax → final disease label

Custom Classifier

- Linear Layer: 1280 → 27 leaf classes
- Global Average Pooling: [1,1280,7,7] → [1,1280]

Weather Risk Analysis

Location Mapping

- City → Latitude & Longitude
- Historical weather API (7 days)

Risk Evaluation

- High risk if:
- Humidity > 80%
 - Rainfall > 20 mm

Response to User

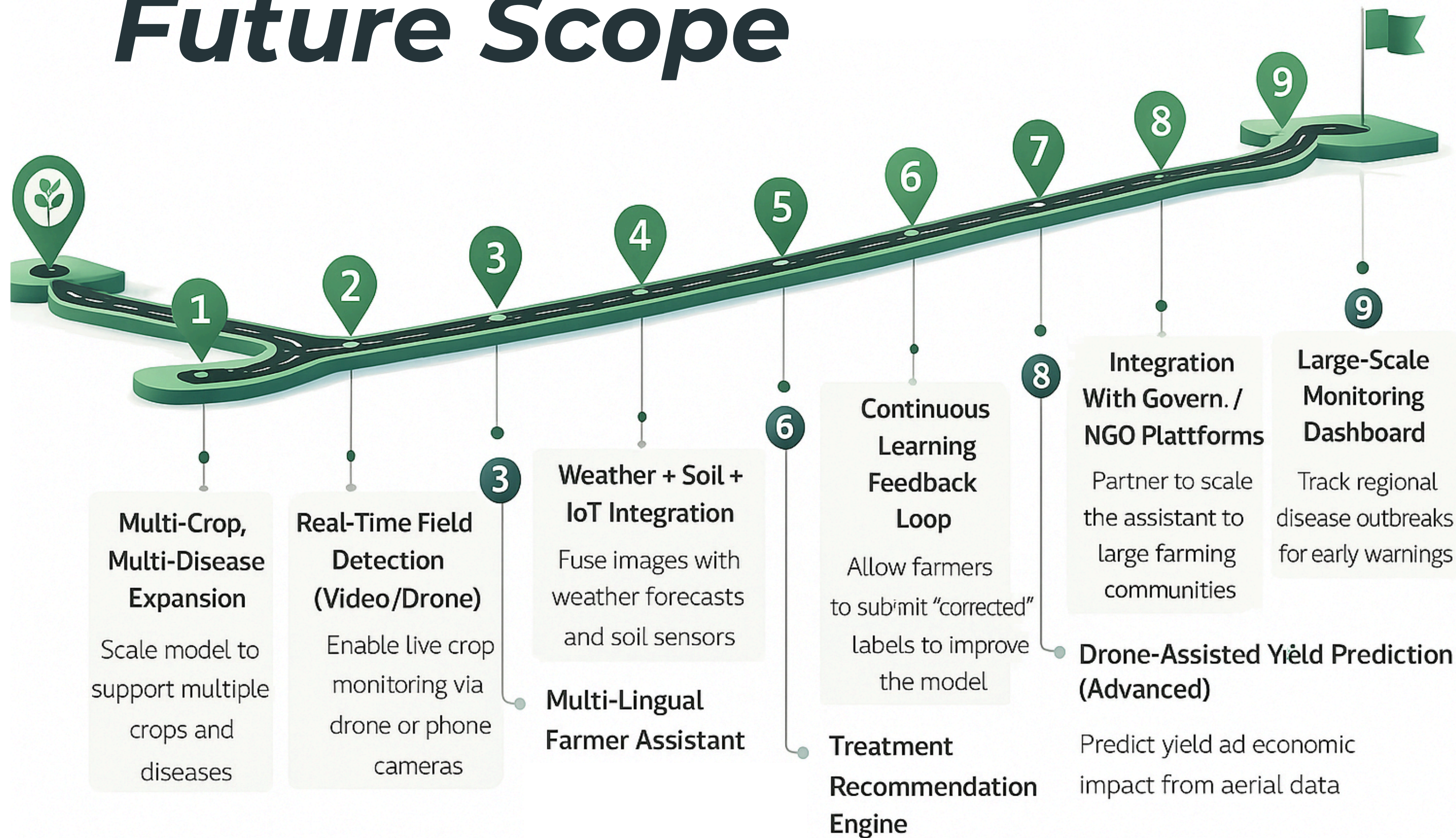
Server Response

- In JSON:
 - Disease label
 - Confidence score
 - Weather-based risk data

Features

- Frontend Visualization
- Confidence bar animation
- Smooth CSS transitions
- Instant feedback to user

Future Scope





MethdAI

Thank You



Project Omicron

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Bibliography

How 3 trillion kg is saved?

- Global food production $\approx$ 8–9 billion tonnes/year
- FAO estimates 20–40% of crops are lost due to pests and diseases
- 30–35% of 8.5 billion tonnes $\approx$ 2.5–3.0 billion tonnes
- 1 tonne = 1000 kg
- $\rightarrow \approx$ 3 trillion kg

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